

Experiment 2 – EE 1200

RC Circuit Response to Step, Pulse, and Square Inputs

IIT Hyderabad

20-Jan-2026

Objective:

To study the time-domain response of an RC circuit for step, pulse, and square-wave inputs and correlate experimental results with hand calculations and simulations.

Part A: Step Input (Single Event)

1. Apply a step input to the RC circuit.
2. Capture charging and discharging waveforms.
3. Extract time constant from measurement.
4. Verify using hand calculations and simulation.

Part B: Pulse Input (Single Event)

1. Apply a single pulse input.
2. Record output waveform.
3. Repeat for multiple pulse widths.
4. Compare results with calculations and simulations.

Part C: Square-Wave Input (Transient and Steady-State)

Case 1: $RC \gg T$

1. Apply square wave with $T \ll RC$.
2. Capture transient and steady-state responses.
3. Compare measurement, calculation, and simulation.

Case 2: $RC \approx T$

1. Apply square wave with $T \approx RC$.
2. Capture transient and steady-state responses.
3. Compare measurement, calculation, and simulation.

Case 3: $RC \ll T$

1. Apply square wave with $T \gg RC$.
2. Capture transient and steady-state responses.
3. Compare measurement, calculation, and simulation.

Deliverables:

- Circuit diagram
- Measured waveforms
- Hand calculations
- Simulation results
- Comparison table
- Brief conclusions